

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Constraint-Based Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO5: Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.
(BL5)

Dave's Taxonomy: Articulation (Level 4)

Question Count: 5

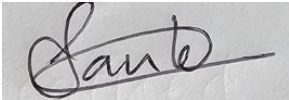
Total Marks: 50

CODE OF CONDUCT

I SANDEEP K / 192521253 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



SANDEEP K / 192521253

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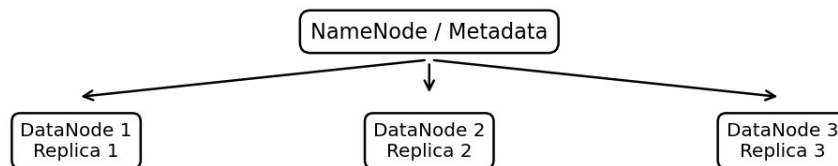
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1. HDFS media storage design

The platform needs high availability with a maximum replication factor of three. HDFS provides distributed storage while controlling the cost of additional copies.

- Use replication factor 3 with rack-aware placement.
- Use NameNode high availability and automatic failover.
- Partition large files by date or media type.
- Apply compression and storage policies where suitable.
- Monitor capacity, replication and rebalancing.

HDFS Storage Architecture



2. MapReduce within 20 minutes

The batch job must finish inside a fixed window, so balanced splits and efficient shuffle processing are essential.

- Create balanced HDFS input splits.
- Run Mapper tasks in parallel for parsing/filtering.
- Use a Combiner where applicable.
- Distribute keys evenly across Reducers.
- Monitor task time and tune parallelism.

MapReduce Workflow



3. YARN scheduler policy

ETL needs priority, but reporting and ML must still receive fair resources. Queue-based scheduling provides predictable allocation.

- Create queues for ETL, reporting and ML.
- Give ETL higher priority/capacity.
- Use fair/capacity scheduling principles.
- Set queue and application limits.
- Enable recovery and rescheduling after node failure.

YARN Resource Management



4. SAN, NAS and database ingestion

Multiple enterprise sources require controlled ingestion and duplicate prevention. Incremental extraction reduces unnecessary processing.

- Use Apache NiFi or ETL connectors.
- Centralize data in HDFS/data lake.
- Use incremental extraction or change tracking.
- Apply record IDs, checksums and validation.
- Maintain independent backups and monitoring.

Enterprise Data Ingestion



5. Secure end-to-end Hadoop solution

The complete platform should secure movement and storage while remaining highly available. Hadoop components can be combined into a controlled processing pipeline.

- Use NiFi/ETL for managed ingestion.
- Encrypt data in transit and authenticate connections.
- Use replicated HDFS with NameNode HA.
- Use YARN for workload management.
- Apply access control, audit, backup and monitoring.

Secure Hadoop Pipeline

